

Mathematics Analysis and Approaches HL Internal Assessment

Correlating Commuter Modal Share with Traffic Congestion Rate in California



Word Count: 3446

1. Introduction

The issue of traffic congestion represents a persistent and costly challenge in modern urban infrastructure, particularly within the United States. My initial observations while residing in the San Francisco Bay Area during the 2019-2020 school year in California revealed the striking paradox of chronic, 24-hour congestion that endures despite substantial public and governmental investment into extensive public transportation systems. This persistence suggests that simply providing transportation alternatives does not guarantee a corresponding decrease in road traffic volume.

This observation serves as the foundation for the present mathematical investigation, which focuses on quantitative solutions to urban traffic mitigation. Urban engineering strategies generally fall into two categories: supply-side solutions (e.g., road expansion and freeway additions designed to increase capacity) and demand-side solutions (e.g., the introduction of dedicated bus lanes, light rail, or policy incentives to promote public transit utilization). Increasingly, modern solutions center on encouraging a modal shift away from single-occupancy vehicles.

Therefore, this exploration aims to investigate the mathematical research question: To what extent does the modal share of non-driving commute methods (Public Transit, Carpool, Walk, and Bicycle, which is considered a mode of transportation in California) correlate with the Aggregate Congestion Rate (ACR) across diverse California Caltrans districts?

I will analyze available data from California, the most populous state in the U.S., which offers a robust and comprehensive dataset due to its dense population distribution and varied, highly developed transportation networks. By examining the correlation between the change in modal share across different commuter methods and established measures of congestion, this work applies mathematical modeling to evaluate the effectiveness of these varying urban planning strategies.

2. Rationale

This investigation was motivated by observing a persistent paradox in the San Francisco Bay Area, California, USA: chronic traffic congestion persists despite massive public investment in alternative commute programs. This observation challenged the assumption that providing non-driving options automatically

reduces traffic volume, driving the need for a quantitative solution to this urban policy dilemma: determining which non-driving commute method correlates most strongly with effective congestion mitigation.

To ensure the analysis is robust, the study uses comprehensive data from all Caltrans districts across California, allowing for a critical comparison between Densely Populated and Not Densely Populated Districts to account for varied urban dynamics.

The core rigor of the analysis lies in the comparative use of the Pearson Product-Moment Correlation Coefficient (PMCC) for linear association and Spearman's Rank Correlation Coefficient for monotonic relationships. The decision to rely heavily on the nonparametric Spearman's correlation due to data size limitations and potential nonlinearity demonstrates a critical, high-level awareness of the appropriateness of the statistical model, which is further validated by the use of P-values for hypothesis testing.

3. Methodology

a. Definition of Terms

According to the interpretation of the definition in the International Encyclopedia of Human Geography 2009, public transportation is defined as such: *“Public transport refers to any mode of transport that is available for hire and reward, and is accessible to the general public.”*

(ScienceDirect *Public transport*). However, modes of transportation such as walking, biking, carpooling, or working from home could also have a significant impact on the congestion rate in theory. We must digress from the general definition and focus on all modes of transportation other than single-person driving to work.

The definition of traffic congestion used in this exploration will be based on the definition “In the transportation realm, congestion usually relates to an excess of vehicles on a portion of roadway at a particular time, resulting in speeds that are slower than normal or 'free flow' speeds.”

Based on the datasets open for public use on [Data.gov.ca](https://data.gov/ca), the official open-source database of the California state government, there are two methods for calculating congestion rates: either by the average speed at stops or on moving sections of the road. Both of the datasets follow similar formulas:

$$Congestion\ rate = \frac{(Speed\ at\ Off-peak\ hours)-(Speed\ at\ Peak\ hours)}{(Speed\ at\ offpeak\ hours)} \cdot 100\%$$

Term	Definition (24hr time)
Off-peak hours	0:00-6:59; 10:00-14:59; 20:00-23:59
Peak hours	7:00-9:59; 15:00-19:59
All-day	0:00-23:59

Table 1: The terms *Off-peak hours*, *Peak hours*, and *All-day*

Note: Data for *All-day* is not used in this investigation.

The definitions are found and classified in the Average Transit Speeds by Route and Average Transit Speeds by Stop datasets published by Caltrans.

The datasets used in this paper will be from the California Open Data Portal website's Caltrans data section (<https://data.ca.gov/organization/caltrans>). Caltrans is the state of California's governmental transportation agency. The entity regularly collects and publishes data from the sensors and satellite updates focused on bridges, highways, and roads in California.

The goal is to calculate the aggregate values of the variables for regression analysis. For this IA, the aggregate value refers to the numerical value for each Caltrans district. This is the weighted average of the values of smaller regions within the Caltrans district, explained further below.



Figure 1: Caltrans district map (*District Map and county chart*)

California's population distribution is notably uneven, primarily due to the state's inherent characteristics and ongoing urbanization trends. This has resulted in a significant disparity, where certain Caltrans regions exhibit exceedingly low population densities, while others are exceptionally densely populated.

This marked divergence in population density suggests a high probability that densely populated districts will demonstrate distinct behavioral patterns and trends when compared to their less densely populated counterparts. Consequently, some of the information presented within this analysis may be exclusively applicable to either "Densely Populated Districts" (referred to as DPD below) or "Not Densely Populated Districts" (referred to as NDPD below). To ensure clarity and precision throughout this article, these two terms, along with the foundational formula for calculating population density, are formally defined as follows:

$$\text{Population density} = \frac{\text{Population (people)}}{\text{Area (sq mile)}}$$

Term	Abbreviation	Definition (districts)
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Densely Populated Districts	DPD	12, 7, 4, 11, 3, 8
Not Densely Populated Districts	NDPD	10, 5, 6, 1, 2, 9

Table 2: Classification of district categories and their abbreviations.

Besides, as shown in the datasets, the state of California considers bicycles, both mechanical and electrical, to be motor vehicles. This is important to take into account since Bicycles will follow motor vehicle regulations and, most importantly, influence the congestion rate.

b. Collection of Data

For this IA, due to the definition of the variables investigated, the “Transportation to Work”, “Average Transit Speeds by Route”, and “Average Transit Speeds by Stop” datasets will be used.

In the Transportation to Work (Referred to as T2W below) dataset, the Excel spreadsheet containing both the data and the data dictionary can be easily downloaded. However, in the Average Transit Speeds by Route (Referred to as *ATSBR* below) and the Average Transit Speeds by Stop (Referred to as *ATSBS* below) datasets, the CSV file does not provide the data dictionary. Therefore, I decided to email Caltrans’s Chief Data Scientist, Dr. Mintu Miah, who suggested I use any trustworthy LLM (Large Language Model) AI such as Google Gemini or ChatGPT to obtain the data dictionary through the `< attrdef >` flag in the metadata XML of the ArcGIS version dataset.

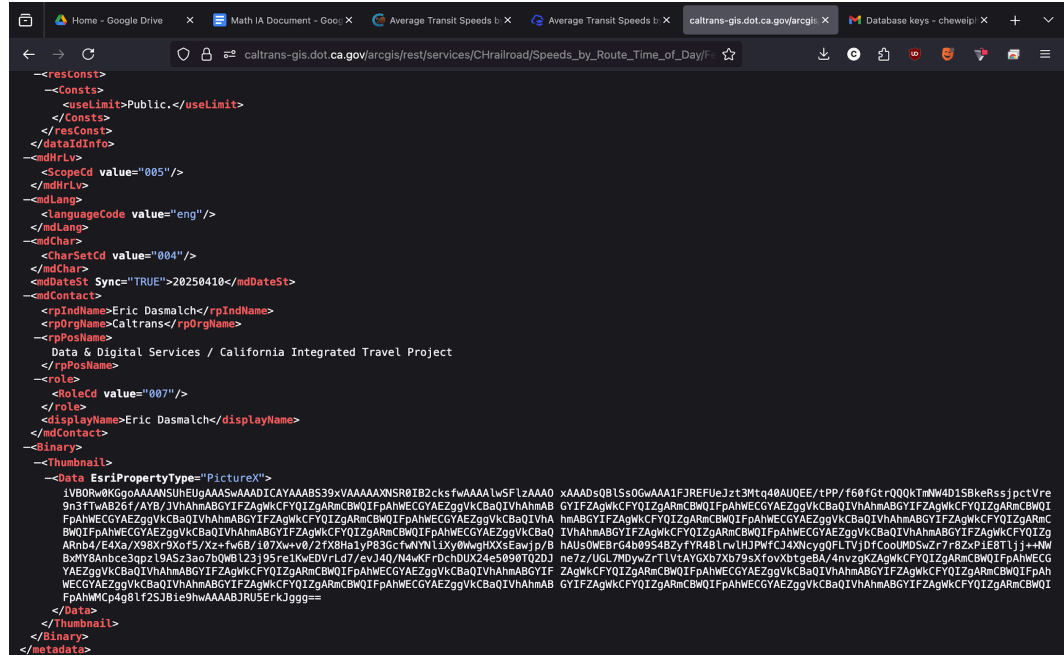


Figure 2: Raw metadata

A critical aspect of this study is the data collection period and aggregation method. While many datasets offer monthly or yearly breakdowns, the Caltrans data used here presents a cumulative average. Caltrans exclusively publishes these averaged figures. These averages are dynamically updated, integrating all data from January 2005 through the present month (June 2025). Consequently, it is imperative to recognize that any results or deductions within this IA are predicated upon traffic values averaged across the entire January 2005 to June 2025 timeframe.

c. Data Cleaning and Initial Processing

Data preparation is essential as the large, publicly available datasets often contain fields irrelevant to this investigation or incomplete entries. Given that three data sets exceed 10,000 rows, manual cleaning is impractical. Therefore, the Python programming language, utilizing the Pandas library, will be employed to efficiently remove extraneous rows and columns and ensure the data's integrity for the subsequent statistical analysis.

In the T2W dataset, the following fields of data need to be removed:

{reportyear, geotypevalue, county_fips, geoname, race_eth_code, geotype, region_name, region_code, ind_id, ind_definition, race_eth_name, LL95CI_percent, UL95CI_percent, percent_se, percent_rse, CA_decile, CA_RR}

In the *ATSB*R dataset, the following fields of data need to be removed:
 $\{route_id, base64_url, org_id, agency, route_name, OBJECTID, time_period, direction_id, district_Shape_Length\}$

In the *ATSBS* dataset, the following fields of data need to be removed:
 $\{route_id, stop_pair, base64_url, org_id, agency, Shape_Length, OBJECTID, time_period, stop_pair_name, direction_id, district_n_trips\}$

The T2W database provides comprehensive data regarding the percentage usage of various transportation modes for commuting to work. To derive a meaningful aggregate rate for analysis within this IA, a specific data processing methodology was implemented. Initially, the dataset was refined by isolating the percentage usage data for a single, selected transportation type per county, while irrelevant data for other modes were removed. This preliminary filtering ensured that only the pertinent information for the targeted analysis remained.

Subsequently, to calculate the aggregate rate of usage for each district, the rate for each county will be disregarded. Instead, the aggregate rate is calculated by dividing the total usage number (*pop_mode* below) of each district by the total population (*pop_total* below).

$$pop_mode = \sum_{county} pop_mode$$

$$pop_total = \sum_{county} pop_total$$

$$district\ rate = \frac{pop_mode}{pop_total} \cdot 100\%$$

Since many traditionally regard driving as the main method of transportation to work. As mentioned above in **Definition of Terms**, this analysis will concentrate on four other transportation methods, identified by their data keys:

$\{CARPOOL, BICYCLE, WALK, PUBLICTR\}$ (representing public transit to work)

Method of Transportation	Rate of Usage (%)
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CARPOOL	25.3
BICYCLE	15.6
WALK	4.18
PUBLICTR	2.15

Table 3: Example data with rates of usage for District 1

The complete calculated rate of usage table is in **Appendix**. Data from District 9 is missing. This is likely due to the convenience sampling of the dataset. While the reason is unknown, it is probably due to District 9 being spread out and large, making data collection difficult.

For both the ATSBR and ATSTS datasets, as per the definition in **Definition of Terms**, the congestion rate is the percentage change in the speed of traffic between *Off-peak* and *Peak*. Since each row of data is based on readings on one sensor, the “aggregate congestion rate” (referred to as *ACR* below) is the weighted average of the congestion rates in each sensor. The weight data ($weight_i$) below is the *Shape_Length* variable for ATSBR and the n_trips variable for ATSTS. The formulas to calculate the congestion rate for each location and the whole district are used as shown below.

$$Congestion\ rate_i = \frac{Off-peak\ speed_i - Peak\ speed_i}{Peak\ speed_i}$$

$$ACR = \frac{\sum_{i=1}^n (Congestion\ rate_i \cdot weight_i)}{\sum_{i=1}^n (weight_i)}$$

District	Average Congestion Rate (ACR)	
	ATSBR (fractional)	ATSTS (fractional)
1	-0.00899	-0.208

Table 4: Aggregate congestion rate for District 1

The full calculated ACR table is shown in **Appendix**.

Following the initial data cleaning phase in this IA, attention will turn to the identification of outliers. The Interquartile Range (IQR) method will be

systematically applied to detect these unusual data points. This technique defines outliers as values significantly deviating from the central distribution, specifically those lying beyond 1.5 times the IQR from either the first or third quartile (Q1 or Q3).

Dataset	Caltrans District	Value
T2W PUBLICTR	4	9.945316
ATSBR	9	-0.2206638
p50 ATSBS	6	-0.411984

Table 5: All outliers

d. Two-variable Regression with Pearson Moment Correlation Coefficient

The core of this process involves examining the statistical correlation between each independent transportation variable and the dependent congestion rate, using the PMCC (r). To ensure the robustness and spatial variability of these relationships, this analysis will be conducted both for the entire dataset (macroscopic, state-wide level) and separately for districts categorized as *DPD*, *NDPD*, and *ALL*. The resulting r value will rigorously evaluate the strength and direction of the linear association between the variables.

Range, r	Correlation
$0 < r \leq 0.25$	Very Weak
$0.25 < r \leq 0.5$	Weak
$0.5 < r \leq 0.75$	Moderate
$0.75 < r \leq 1$	Strong

Table 6: The interpretation of different r values.

The regression operation will be conducted using Microsoft Excel's Data Analysis Toolpak, specifically leveraging its scatter plot generation and dedicated "Regression" feature. Beyond calculating the r , the Toolpak provides the P-value (identical to "*Significance F*" in two-variable

regression), which will be used to test the statistical validity of the correlation. A P-value greater than 0.05 (5%) suggests the observed correlation is likely due to chance (supporting the null hypothesis, H_0), thus signifying that the study's hypothesis of a correlation (H_1) may be false.

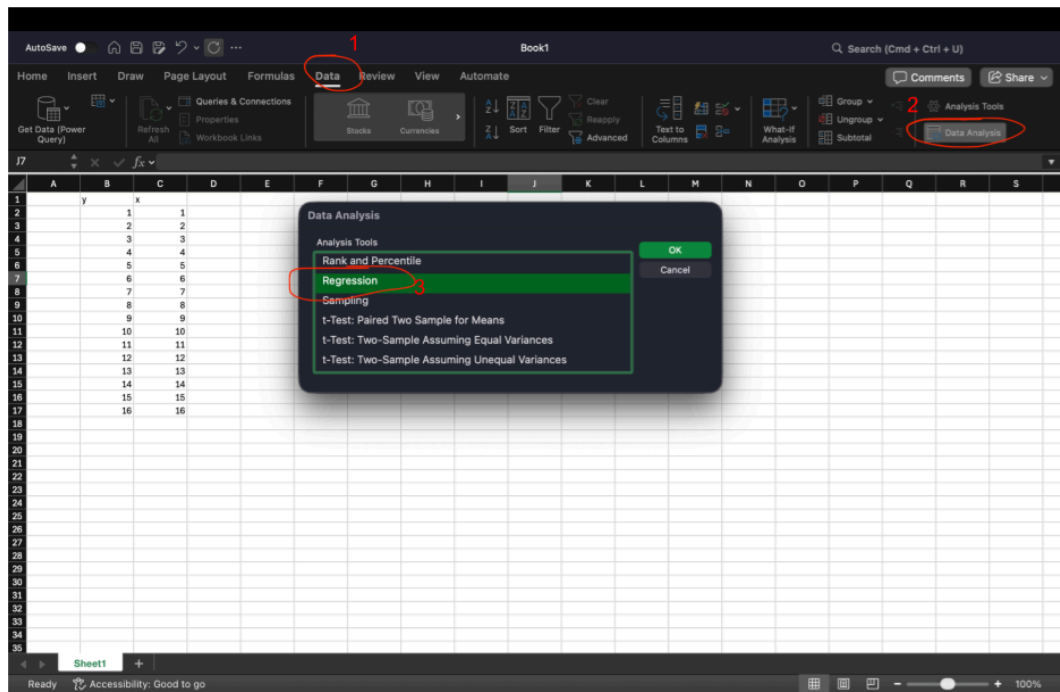


Figure 3: Microsoft Excel’s Data Analysis Toolpak

The variable pairs in the two-variable regression are listed in the table below.

Dependent Variables	ATSBS, ATSBR
Independent Variables	PUBLICTR, CARPOOL, WALK, BICYCLE

Table 7: Independent and dependent variables for two-variable regression with PMCC

e. Multi-variable Regression

Following the initial bivariate analysis, we move to multivariable regression to build a more robust predictive model of traffic congestion (ATSBS or ATSBR). The goal is to evaluate the combined influence of all four

transportation variables. The same analysis ToolPak will be used. However, due to data limitations and a small sample size, Microsoft Excel cannot successfully run the full regression. As results would be unreliable (high P-value, low significance), we must limit the analysis to calculating the correlation using only two independent T2W variables at a time. See **Appendix** for data.

Dependent Variables	ATSBS, ATSBR
Independent Variable Pairs	(PUBLICTR, CARPOOL), (WALK, BICYCLE)

Table 8: Independent and dependent variables in the multiple-variable regression

f. Two-variable Regression with Spearman's Rank Correlation Coefficient

After the initial bivariate and multivariable regression attempts yielded unreliable results (see **Analysis** for full explanation), consultation with my peers who take IB math AI and transportation engineering expert Professor Dr. Ming-Heng Wang, led to the decision to investigate Spearman's rank correlation coefficient (ρ or r_s) into all the sets.

Unlike PMCC, which uses raw data to measure strictly linear relationships, Spearman's coefficient is calculated based on the ranks of the variables. The value is actually the PMCC of the rank of the data. This method is superior because it accurately assesses monotonic (non-linear) relationships. Furthermore, by using ranks instead of raw values, r_s is significantly less sensitive to outliers or skewed distributions (a known issue with the PMCC results), providing a more reliable measure of correlation for this dataset.

The formula for Spearman's Rank Correlation Coefficient is written below:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

ρ : Spearman's rank correlation coefficient

d_i : difference between the two ranks of each observation

n : number of observations

Just like the Pearson Moment Correlation Coefficient, Spearman's rank correlation coefficient for each set will be calculated using Microsoft Excel spreadsheets.

4. Analysis

a. Two-variable Regression with Pearson Moment Correlation Coefficient

After performing two-variable regression with Pearson Moment Correlation Coefficient (data in **Appendix**), unexpectedly, only five sets of regression results out of 24 were sufficient to conclude correlation. The five sets and their PMCC (r) are listed in the table below.

Congestion rate measure	Mode of transportation	Specification	r
ATSBS	PUBLICTR	NDPD	-0.996
ATSBS	WALK	ALL	-0.734
ATSBR	PUBLICTR	ALL	0.663
ATSBR	CARPOOL	ALL	0.580
ATSBR	BICYCLE	ALL	0.594

Table 9: Sets with r values of significance

The data analysis was severely limited, with only 11 districts providing sufficient data, resulting in most correlations having no statistical significance (high P-value). For example, the ATSBP-PUBLICTR (*NDPD*) set, despite a perfect ' r ' value, was deemed unreliable due to severe over-fitting with only four data points. However, four sets showed statistically significant trends based on the specific context of California's non-urban districts and different congestion metrics (ATSBP for intersections and ATSBP for open roads).

The four significant correlations challenge conventional beliefs about commute methods:

- i. Walking (ATSBP vs. WALK)

Walking showed a strong negative correlation with intersection congestion (ATSBS-WALK) because pedestrians force cars to slow down at corners. Conversely, on open roads (ATSBR-WALK), cars simply circumvent pedestrians, leading to no significant trend.

- ii. Public Transit (ATSBR-PUBLICTR)
Public transit showed a moderate positive correlation (increased congestion). In non-urban areas, buses often obstruct traffic without significantly reducing the volume of private cars, making transit a burden.
- iii. Carpooling (ATSBR-CARPOOL)
Carpooling also showed a moderate positive correlation, suggesting that the extra distance required for pick-up and drop-off in spread-out areas induces more congestion rather than relieving it.
- iv. Bicycles (ATSBR-BICYCLE)
The usage of bicycles displayed a moderate positive correlation. Since California classifies bicycles as motor vehicles, their slower speeds and unpredictable movements on roads without dedicated lanes force cautious drivers to slow down, increasing measured congestion.

b. Multi-variable Regression

After performing the multiple variable regression explained in **Methodology**, despite reducing the number of independent variables to two, no results were found due to the high P-value.

c. Two-variable Regression with Spearman's Rank Correlation Coefficient

Congestion rate measure	Mode of transportation	Specification	ρ
ATSBS	PUBLICTR	all	-0.3666667
		crowded	-0.5

	WALK	uncrowded	-1
		all	-0.6121212
		crowded	-0.7142857
	BICYCLE	uncrowded	-0.8
		all	0.25
		crowded	0.4
	CARPOOL	uncrowded	-0.8
		all	0.2
		crowded	0.37142857
		uncrowded	-0.8
ATSB	PUBLICTR	all	0.61212121
		crowded	0.8
		uncrowded	0
	WALK	all	-0.3090909
		crowded	-0.3714286
		uncrowded	-0.6
	BICYCLE	all	0.46363636
		crowded	0.37142857
		uncrowded	0.2
	CARPOOL	all	0.39090909
		crowded	0.5
		uncrowded	0.95382097

Table 10: Two-variable Regression with Spearman's Rank Correlation Coefficient

i. Public Transportation (PUBLICTR)

Public Transit shows complex results, highlighting the distinction between correlation and causation. When correlated with ATSB, crowded districts (*DPD*) show a strong positive correlation ($\rho = 0.8$). This result means districts with high transit usage also rank high in mid-block slowdowns. This is interpreted as transit being a symptom of pre-existing congestion: high-capacity transit is implemented on major intersections that are already chronically overloaded.

Conversely, when PUBLICTR is correlated with ATSB, uncrowded districts (*NDPD*) display a perfect negative correlation $\rho = -1.0$. This demonstrates that where baseline traffic is low, the marginal increase of public transit perfectly prevents the growth of general congestion at intersections. However, as explained above, this high value is likely due to data overfitting with a small sample size and should be treated cautiously, drawing no definitive mathematical conclusion.

ii. Walking (WALK)

Walking, representing the most direct vehicle substitution, shows the most reliable, consistently moderate to strong negative correlation with ATSB. The correlation peaks at $\rho = -0.714$ in *DPD* and $\rho = -0.8$ in *NDPD*. This consistency confirms that walking successfully eliminates short-distance car trips that typically involve intersection movements and queuing, reliably reducing the overall time spent below the posted speed limit at these critical nodes.

However, this strong correlation significantly diminishes against ATSB. The Walk share correlation drops to a weak negative $\rho = -0.371$ in *DPD*. This finding indicates that while walking relieves congestion caused by local trips at intersections, its impact on high-speed, high-volume, mid-block flow is mathematically marginal, suggesting that mid-block congestion is driven by other non-substitutable factors like long-distance commuting or freight volume.

iii. Bicycle (BICYCLE)

The Bicycle modal share analysis reveals a distinct profile, especially when measured against ATSB. The Bicycle share shows a weak positive correlation $\rho = 0.464$ across all districts. This indicates that the districts with higher cycling rates are also those that experience more mid-block slowdowns. This is interpreted as cycling infrastructure (e.g., dedicated bike signals) often being prioritized for high-traffic intersections, thus correlating with pre-existing mid-block congestion, rather than the cycling rate itself causing the slowdowns.

Against ATSB, the correlation remains weak (e.g., $\rho = 0.4$ in *DPD*). This suggests that while cycling is a vehicle substitute, its overall volume is likely too low to produce the consistently strong negative correlation observed in the Walking share, failing to meaningfully reduce congestion across districts.

iv. Carpool (CARPOOL)

The correlation between Carpool usage and ATSB varies significantly by density. In *DPD*, the correlation is weakly positive $\rho = 0.371$, suggesting carpooling increases congestion at busy intersections. However, in *NDPD* districts, the relationship is a strong negative correlation $\rho = -0.8$. This indicates that where traffic volume is low, carpooling significantly reduces the total vehicle count entering an intersection, successfully preventing the buildup of general congestion. In high-volume areas, the benefit of having two people in one car is largely negated by the total sheer volume of vehicles waiting at signals.

Carpool modal share generally shows a moderate-to-weak relationship with most metrics, except for a major outlier: its correlation with ATSB in *NDPD*, which registers a very strong positive correlation $\rho = 0.954$. Given that this subset of the data contains a relatively higher number of districts compared to the small samples that resulted in suspected overfitting, the resulting coefficient $\rho = 0.954$ can be treated as a reliable mathematical finding. This strong positive rank correlation indicates that high carpool usage is a feature of long-distance traffic that is inherently prone to periodic, severe slowdowns along the roadway, suggesting that the underlying commute structure, rather than the act of carpooling itself, dictates the mid-block congestion profile. Districts

with high carpooling are typically those with long-distance commutes, often utilizing freeways.

5. Conclusion

The primary goal of this investigation was to determine the correlation between non-driving modal share and the Aggregate Congestion Rate. Due to data constraints, Spearman's correlation was used as the reliable non-parametric measure, confirming that the relationship is complex and dependent on the commute method and population density.

The strongest negative correlation was the Walk modal share. This supports prioritizing walking infrastructure to mitigate localized queuing. Other methods proved contextual: Carpooling was negatively correlated only in uncrowded districts (NDPD), but its positive correlation with mid-block congestion suggests it is a feature of long-distance freeway commutes, not a mitigator there. Public Transportation's high positive correlation suggests reverse causality that transit installation is a response to, rather than a primary solution for, chronic congestion. Bicycles showed consistently weak correlations.

In conclusion, modal shift correlates with congestion, but its impact is specific. Mathematically validated strategies should focus on promoting walking in high-density areas and carpooling in low-density corridors.

6. Reflection

While this investigation provides a foundational understanding of the relationship between modal share and congestion, several avenues for mathematical refinement exist to enhance the robustness and generalizability of these conclusions.

One primary area for improvement involves redefining the congestion rate. While this model utilizes the percentage speed difference to calculate the congestion rate, future studies could adopt more multifaceted metrics, such as the Travel Time Index or Delay per Capita, to provide a more nuanced view of the commuter experience, potentially yielding different correlation strengths.

To move beyond simple correlation toward a causal understanding, a multivariate regression model should also be employed. Accounting for confounding variables

such as population density, fuel prices, and last-mile connectivity would allow for the calculation of partial correlation coefficients, isolating the specific impact of modal share on road networks.

Furthermore, expanding the temporal scope to include *All-day* datasets would determine if the observed correlation holds during off-peak hours. This would identify whether the impact of public transit is a systemic reduction in baseline congestion or if it is localized strictly to peak rush-hour periods.

To test the mathematical universality of these findings, the model should be applied across diverse urban archetypes. Comparing Spearman's Rank coefficients in high-density European cities versus sprawling US regions could reveal how infrastructure maturity and urban design influence the effectiveness of modal shifts.

Increasing the sample size and data granularity would allow for the application of the Central Limit Theorem and the exploration of non-linear regression models. Logarithmic or exponential fits might better represent the "law of diminishing returns" often associated with urban transit adoption, transitioning this model from a descriptive tool to a predictive one for urban planning.

Ultimately, while the "Walk" modal share showed the most reliable negative correlation, the positive carpooling correlation in low-density areas provided the greatest intellectual stimulus. To extend this work, my next step would be to utilize a multivariable linear regression model on a larger dataset to overcome sample limitations and build a genuinely predictive mathematical formula for congestion mitigation.

7. Bibliography

1. "California's Population Drain." Stanford Institute for Economic Policy Research (SIEPR), siepr.stanford.edu/publications/policy-brief/californias-population-drain#:~:text=California%20is%20still%20the%20largest,percent%20of%20the%20U.S.%20population. Accessed 20 June 2025.
2. "Public Transport." Public Transport - an Overview | ScienceDirect Topics, www.sciencedirect.com/topics/social-sciences/public-transport. Accessed 23 June 2025.
3. Transportation to Work - Dataset - California Open Data - ca.Gov, data.ca.gov/dataset/transportation-to-work. Accessed 11 July 2025.

4. "Average Transit Speeds by Route." California Open GeoPortal, data.ca.gov/dataset/average-transit-speeds-by-route. Accessed 11 July 2025.
5. "Average Transit Speeds by Stop." California Open GeoPortal, data.ca.gov/dataset/average-transit-speeds-by-stop. Accessed 11 July 2025.
6. District Map and County Chart, cwwp2.dot.ca.gov/documentation/district-map-county-chart.htm. Accessed 16 July 2025.
7. Tu, Anh Trinh, and Thi Phuong Linh Le Le. "Encouraging Public Transport Use to Reduce Traffic Congestion and Air Pollutant: A Case Study of Ho Chi Minh City, Vietnam - Sciencedirect." Encouraging Public Transport Use to Reduce Traffic Congestion and Air Pollutant: A Case Study of Ho Chi Minh City, Vietnam, 14 Mar. 2016, www.sciencedirect.com/science/article/pii/S187770581600401X.
8. California. Vehicle Code, § 21200 (2024).

Appendix

a. Rate of Usage of the Method of Transportation to Work

District	CARPOOL (%)	BICYCLE (%)	WALK (%)	PUBLICTR (%)
1	25.3	15.6	4.18	2.15
2	11.0	0.529	2.78	0.747
3	39.3	28.3	3.68	2.27
4	12.6	3.19	3.60	9.95
5	24.9	13.1	4.26	3.15
6	23.7	8.70	2.65	1.33
7	44.7	33.6	2.75	4.72
8	34.0	19.3	2.38	1.60
10	20.7	7.60	2.24	1.35

11	10.7	0.605	2.83	3.20
12	49.6	39.2	2.49	2.49

b. Average Congestion Rate (ACR)

District	ACR (fractional)	
	ATSBR	ATSBS
1	-0.00899	-0.208
2	-0.0708	-0.133
3	0.0191	-0.150
4	0.0306	-0.138
5	-0.0136	-0.266
6	0.00864	-0.412
7	0.0558	-0.142
8	0.0193	-0.132
9	-0.221	-0.367
10	-0.00778	-0.155
11	0.0335	-0.159
12	0.0553	-0.137

c. Two-variable Regression with Pearson Moment Correlation Coefficient

Congestion rate measure	Mode of transportation	Specification	r
ATSBS	PUBLICTR	all-day	-0.2271516

		crowded	-0.299096
		uncrowded	-0.9963029
	WALK	all	-0.7343421
		crowded	-0.3356904
		uncrowded	-0.8606023
	BICYCLE	all	0.19896347
		crowded	0.61353072
		uncrowded	-0.8083031
	CARPOOL	all	0.12393194
		crowded	0.42476379
		uncrowded	-0.8055475
ATSBR	PUBLICTR	all	0.66344428
		crowded	0.69094472
		uncrowded	0.41970175
	WALK	all	-0.1805113
		crowded	-0.411181
		uncrowded	0.09790708
	BICYCLE	all	0.5943633
		crowded	0.49243636
		uncrowded	0.71377141
	CARPOOL	all	0.5796927
		crowded	0.99023242
		uncrowded	0.42538948

d. Multi-variable Regression

Congestion rate measure	Mode of transportation	Specification	coeff. 1	coeff. 2	const.
ATSBS	PUBLICTR, CARPOOL	all	-0.0040	0.0005	-0.1586
		crowded	-0.0029	0.0005	-0.1532
		uncrowded	-0.0830	0.0029	-0.1023
	BICYCLE, WALK	all	0.0001	-0.0220	-0.0862
		crowded	0.0002	-0.0042	-0.1343
		uncrowded	-0.0040	-0.0107	-0.1010
ATSBR	PUBLICTR, CARPOOL	all	0.0143	0.0014	-0.0647
		crowded	0.0104	0.0005	-0.0102
		uncrowded	-0.0171	0.0066	-0.1277
	BICYCLE, WALK	all	0.0016	-0.0071	0.0084
		crowded	-0.0085	0.0004	0.0521
		uncrowded	-0.0300	0.0073	0.0117